# FLEET TEAM DIGEST

## 737MAX-FTD-27-17004

**Issue Title :** Horizontal Stabilizer Trim Actuator Ball Screw Phosphate Coating Wear Markings

| Airplane Model | ATA | Minor Model(s) |
|---|---|---|
| 737MAX | 2741-81 | -7, -8, -8200, -9, -BBJ |

| Other Model(s) | ECCN |
|---|---|
| 737NG | 9E991 |

| Originated Date | Last Revised Date | Created On |
|---|---|---|
| 06/05/2017 | 06/05/2017 | 04/11/2017 |

| Estimated Completion Date | Next Update Date |
|---|---|
| | |

## Applicability

All 737MAX airplanes

## Description

A 737NG operator reported that while performing the detailed visual inspection of the Ballscrew per the instructions included in Boeing Alert Service Bulletin 737-27A1277, portions of the Phosphate Coating on the threadforms of the Horizontal Stabilizer Ballscrew Shaft had worn off over time. The mechanic originally reported finding corrosion on the ballscrew, but further examination of the ballscrew, revealed that portions of the Phosphate coating had worn off. An example of the reported Phosphate Coating wear can be seen in the attached pictures to this article.  For 737 MAX aircraft, this visual inspection instruction is contained in Aircraft Maintenance Manual Task 27-41-81-210-801.

## Status

Phosphate coating is applied to the threaded surface of the ballscrew shaft during production and acts along with the grease as a "corrosion inhibitor". The phosphate coating does not protect the ballscrew from corrosion by itself. The phosphate coating is porous. Grease (oil) penetrates into the pores and the combination of two will protect the Ballscrew from corrosion. The phosphate coating on the Ballscrew is approximately 0.0002”-0.0004” thick.

The Umbra Cuscinetti S.p.A. Ballnut incorporates rubber wiper seals that are designed to prevent contamination from entering the Ballnut during actuation. When new, these seals interface very tightly with the screwshaft and can actually "wear" the phosphate coating if the lubricant film on screwshaft is permitted to dry.

Under normal actuator operation, there will be a continuous light film of lubricant on the ballscrew threads which will decrease the friction between the seal and ballscrew shaft. Additionally, these rubber seals will wear in service to reduce the preload between the seal and the phosphate coating.

## Interim Action

None

## Final Action

Corrosion of the screw will be reddish brown (or whatever color is technically correct for describing rust related with the corrosion of steel) while discoloration of the screw in different tones of grey, silver and black is not corrosion but something

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related to the application and/or wear of the phosphate coating. The latter is not detrimental to continued operation of the HSTA so long as the lubrication events continue to occur at least every 1,600 hours of service.

## Reference Categories

| Reference Type | Reference Number | Revision Number | Reference Date |
|---|---|---|---|
| Aircraft Maintenance Manual (AMM) | 737MAX - D633AM101: Task 27-41-81-210-801 | | 05/15/2017 |
| Fleet Team Digest Article (FTD) | 737NG-FTD-27-12003 | | 12/09/2014 |

## Parts List

| Part Name | Boeing Part Number | Supplier Name | Supplier Number |
|---|---|---|---|
| Ballscrew | 10-62210-2/-3/-4/-5/-6/-7 | Umbra Cuscinetti S.p.A. | 07322P000-01/-02/-03/-0 |

## Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- In Service Occurrences : null

## Attachments:

a) Phosphate Coating Wear - Figure 2

b) Phosphate Coating Wear - Figure 1

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